Performance Evaluation of DigiMesh and ZigBee Wireless Mesh Networks

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***Abstract*—This paper presents an experimental evaluation of ZigBee and DigiMesh networks; a two well-known and widely used Wireless Mesh Networks architectures. The performance metrics used were the throughput, Round Trip Time, Received Signal Strength Indication, and Mesh Routing Recovery Time. The conducted experiments show that DigiMesh have better throughput than ZigBeee networks. However, the later outperforms DigiMesh based networks by having lower Round Trip Time, higher Received Signal Strength Indication, and it needs less time to recover from a failure node. These features make ZigBee a better choice for applications that require less delay and need longer communication range, while DigiMesh based networks are better used for networks with high throughput requirements.**

***Index Terms*—Wireless Mesh Networks, ZigBee, 802.15.4, DigiMesh.**

I. Introduction

A wireless mesh network (WMN) is a communication network that consists of radio nodes, organized in a mesh topology where there is no central node that all the traffic should go through, and all nodes can send directly send to each other, which not only makes the network scalable, but also eliminate the risk of having a single point of failure, which is the case in centralized networks. In mesh networks, this is achieved by having several redundant links and paths to the destination node, so if one of these node fails, then other nodes can be used to reroute the data until it reaches to the destination sink node. Thus, wireless mesh networks can adapt to network changes and selfheal from failures. WMS can be implemented with various wireless technologies such as 802.11, 802.15, 802.16, cellular technologies and need not be restricted to any one technology or protocol [1]. Furthermore, Mesh networks are very scalable in terms of being able to send data over large distances by utilizing the intermediate nodes, which act as routers in a multi-hop network, without the need of having long-range wireless communication module. Such a multi-hop network architecture may, with careful design, provide high bandwidth, high range, low power communication capability, low interference and high spectral utilization [1].

In this type of network infrastructure. Nodes act as routers to transmit data from nearby nodes to other nearby nodes until the data reach the final destination. Further, mesh network are immune to nodes' failure which may occur if one node drops out of the network, due to hardware failure or any other reason, then the neighbors can quickly find another route using a routing protocol such as the Ad-hoc On-demand Distance Vector-Routing Protocol (AODV) [1]. WMNs have several applications especially when used with sensors to remotely monitor the area of interests, and send notification or alarms in case detecting any abnormal event. Some of these examples are in border monitoring and security, smart grids, structure health monitoring, remote patient monitoring, smart homes, etc.

In the literature, several works have been presented that described the implementation details for various WMN based applications. For instance, the authors in [2] presented how WMNs can be used in a smart home system to provide remote connection for various home-based IoT devices, the authors utilized DigiMesh technology [7] for building the mesh network. In [3]. the authors presented the design and implementation details of AquaMesh, a system for monitoring several aqua-environmental parameters, where different wireless technologies such as ZigBee, WiFi and GPRS are used to send an alert message if certain threshold value of these parameters are exceeded. In the healthcare field, a blood pressure monitoring system for hemodialysis patients is proposed in [4]. The authors proposed to used ZigBee mesh network to send the blood pressure values for the patient under monitoring, where they are stored in their health records for analysis. Further, an alarm message will be sent to the healthcare personnel in case of detecting any abnormal situation. In structure health monitoring systems, the authors in [5] presented a multi-function interferometric radar sensor with ZigBee mesh-networking functionality, where the structure displacement is monitored and alter message is sent accordingly.

The implementation aspects of ZigBee WMN was presented in [6], where the technical details of building a WMN utilizing ZigBee technology with its performance evaluation is presented. However, the paper does not show experimental results on all the vital transmission performance metrics such as throughput, delay, time to recover from node failure, and Received Signal Strength Indication (RSSI). Further, it does not extensively focus on the transmission range validation and limitation of this protocol. Further, a comparison study between ZigBee and DigiMesh has be presented in [7]. However, the white technical paper did not show any performance testing or

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comparisons between the two networks. Furthermore, detailed steps to configure a WMN using different configuration modes have been presented in [1]. However, the book did not include any performance testing or comparisons with different WMN technologies. The aim of this paper is to focus on the WMN performance evaluation, it also compares the performance with two widely used and well known technologies, ZigBee and DigiMesh, where the pros and cons of each technology is presented after conducting a set of experimental tests. The paper is a continuation of our previous efforts that were focused on proposing protocols and mechanisms for saving power [8–11] and reducing interference [12,13]. The rest of the paper is organized as follows: Section II provides a brief description about the IEEE 802.15.4, the ZigBee and DigiMesh protocols. Section III describes the hardware and configuration used to build the WMN. Section IV presents the experimental performance evaluation. Finally, Section V concludes the paper and proposes future work.

II. WMN Protocols

In this section, a brief overview on the protocols that are widely used in WMN is presented. In particular, the IEEE 802.15.4 physical layer protocol, the ZigBee and the DigiMesh MAC protocols are presented.

A. IEEE 802.15.4

802.15.4 is an IEEE standard for wireless communication by that has been developed to meet the needs of the applications that require lower data rate, simple connectivity and low battery consumption. The 802.15.4 standard specifies that communication can occur in the 868–868.8 MHz, the 902–928 MHz or the 2.400–2.4835 GHz Industrial Scientific and Medical (ISM) bands. Despite the fact that from technical point of view, any of these bands can be used by 802.15.4 devices, the 2.4 GHz band is more popular as it is open in most of the countries worldwide. The 868 MHz band is open for European use, whereas the 902–928 MHz band can only be used freely in the United States, Canada and a few other countries.

The 802.15.4 standard specifies that communication should take place in 5 MHz channels starting from 2.405 to 2.480 GHz. Further, the standard specifies 250 Kbps maximum over the air data rate, but due to the protocol overhead, the actual data rate is approximately half of that. The protocol utilizes the Direct Sequence Spread Spectrum (DSSS) and an Offset Quadrature Phase Shift Keying (O-QPSK) with half-sine pulse shaping modulation scheme, which provides efficient interference mitigation and immunity. The 802.15.4 protocol is widely used in WMN and forms the basis for many MAC protocols such as 6LowPan, Thread, ZigBee, and DigiMesh. In what follows, a brief description is provided on the two later protocols that are used in this paper.

B. ZigBee

ZigBee protocol is designed to address the needs of having low power; low data rate specifications needed in various

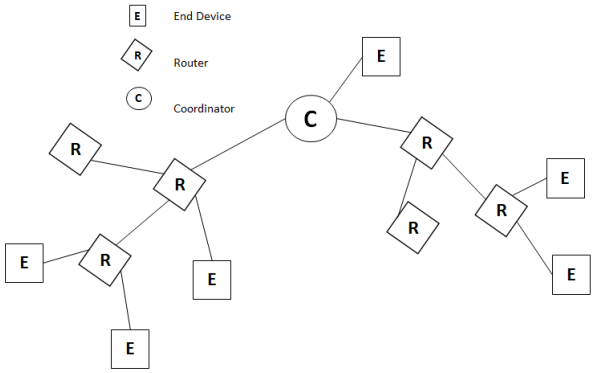

Fig. 1. Node types in a ZigBee network [7].

IoT devices. It is led by the ZigBee Alliance[1] with more than 300 members.[2] ZigBee has attracted attention due to its open architecture, which facilitates vendor interoperability. Further, it allows the existence of low cost end devices as will be explained later, which facilitates its adoption. ZigBee runs on top of 802.15.4 protocol to provide additional routing and networking functionality. One key advantage of ZigBee that it can easily provide mesh networking for the underlying 802.15.4 protocol. The ZigBee protocol defines three types of nodes: Coordinators (C), Routers (R) and End Device (E), with a requirement of one Coordinator per network. All nodes are capable of sending and receiving data. However, there are some differences in their specific functionalities, as will be explained below.[3]

- Coordinators are the most vital nodes among the three nodes types. For each ZigBee network, there should be one coordinator that stores the network information, generate the network beacon signal, and stores the network security keys [1].
- Routers nodes work as intermediate nodes that are responsible for relaying data from other devices [1].
- End Devices are the lowest capable nodes that can directly communicate with its parent node, which could be either a router or a coordinator node. As such, they do not perform any routing functionality and they could have low battery capabilities, which makes them cost effective and have low requirements [1].

C. DigiMesh

DigiMesh is a proprietary protocol that was developed by Digi Inc.,[4] which is one of the pioneers in the field of Internet of Things and machine-to-machine communication and networking. DigiMesh is widely used and adopted in several manufacturers and are used in various applications [7]. DigiMesh consists of one router node, as such, the network is

[1] http://www.zigbee.org/.

[2] "XBee 802.15.4 Protocol Comparison" Web. <http://www.digi.com/pdf/xbee-802-15-4-protocol-comparison>.

[3] "Wireless Mesh Networking ZigBee Vs. Digimesh" Web. <http://www.digi.com/pdf/wp_zigbeevsdigimesh.pdf>.

[4] http://www.digi.com.

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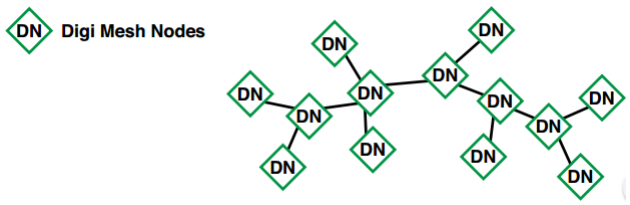


Fig. 2. Node type in a DigiMesh network [7].

simple and homogeneous, where all nodes are interchangeable and can route data. There is no parent-child relationships as described in ZigBee, where the coordinator node is the parent for the router or End Device nodes, and the router nodes are the parents of other router or End Device nodes.

This network heterogeneity makes the network setup and establishment simple, flexible, extensible, and reliable. Further, it uses a simplified addressing method, which improves network setup and trouble shooting. The message could be transmitted through routers, which act as relay to reach the destination. As a result, the ability to relay messages allows the network to maximize range while also allowing a meshing capability [7]. DigiMesh has higher frame payload than ZigBee protocol, which improves the throughput. A characteristic that is needed by applications that send data with large block size. More technical details and comparisons with ZigBee can be found in [7].

III. Wireless Mesh Network Implementation

The aim of this section is to describe the basic hardware and software building blocks needed to establish a WMN.

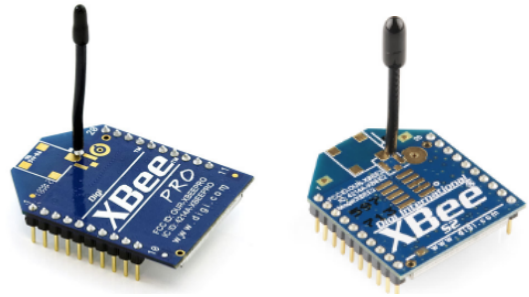
A. Hardware

XBee is a wireless communication module manufactured by Digi Inc. that can be used to build a WMN. It can run either ZigBee or DigiMesh protocols depending on its type, and by utilizing the appropriate firmware. XBee modules are characterized by having low-power, low-cost applications. Another version used for higher power and longer-range applications is the XBee-Pro modules. Both modules are configured using the XCTU software provided by Digi[5] and can be used to establish ZigBee WMN. The modules used to build the DigiMesh network are the Series 1 XBee Pro modules. However, to build a DigiMesh network, their firmware should be upgraded to support DigiMesh protocol. To build the ZigBee network, Series 2 modules were used, which only supports the ZigBee protocol.

B. Wireless Mesh Network Configuration

In general, there are three interaction modes with XBee modules, which are AT commands, transparent, and API. These three interaction modes apply only to interactions that take place via the XBee's serial connection, also known as the UART. Wireless communication between XBees are

[5] http://www.digi.com.



(a) (b)

Fig. 3. (a) Series 1 XBee Pro 60 mW module, (b) Series 2 XBee 2 mW module.

independent of the chosen interaction mode. Therefore, a radio configured to operate in transparent mode may communicate wirelessly with another radio configured in API mode. In this paper, all the three types of configurations have been utilized. The detailed steps to configure a WMN using these modes are fully explained in [1], which were not included in the paper due to its size limitation.

IV. Perfoamnce Evaluation

In this section, a performance evaluation of the DigiMesh and ZigBee WMN is presented. The performance metrics used are: the Received Signal Strength Indicator, packet delay, throughput and mesh routing recovery time. The results of the study are important for understanding the performance of the wireless sensor network where the communication distance between the nodes is greater than a point-to-point communication link, and hence communication requires multi-hop transmissions.

A. Range of Communication

A range test was carried out to measure the maximum possible distance between any two nodes before communication is lost. The wireless sensor networks built earlier were used in the tests. For both the DigiMesh and ZigBee networks, the position of the coordinator node was fixed and the End Device node was moved until the communication was lost. For DigiMesh network, the distance was found to be around 40 m. As for the ZigBee network, the distance was found to be 120 m. Therefore, using ZigBee network, much greater ranges could be achieved across the entire network.

B. Received Signal Strength Indicator

RSSI is the signal strength level of a wireless device of the last good packet received and is measured in dBm. In free space, the Received Signal Strength (RSS) decays with distance travelled away from the sender. The ratio of the received power P_r to the transmission power can be expressed as follows using Friis transmission equation [1]:

$$P_r = P_t \cdot G_r \cdot G_t(\lambda/4\pi d)^2 \quad (1)$$

where G_t and G_r are the gains of the transmitter and the receiver antennas, respectively, and d is the distance between

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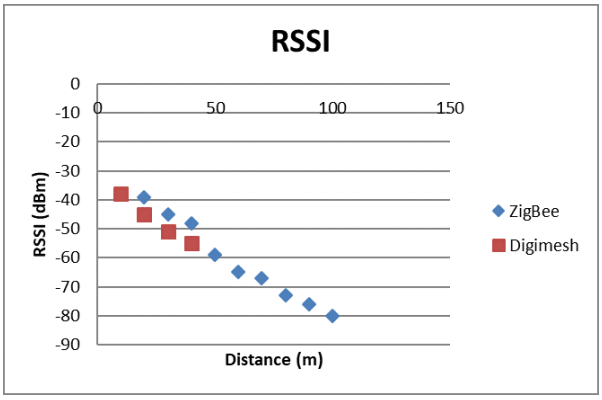

Fig. 4. RSSI strength test versus the distance.

the sender and the receiver. From the equation above, it can be seen that as the distance increases, the power received decreases. RSSI can be defined as the ratio of the received power P_r to the reference power P_{ref} as shown in Eq. (2) [1].

$$\text{RSSI} = 10\log(P_r/P_{ref}). \tag{2}$$

Fig. 4 shows how the values of measured RSSI are less for the DigiMesh network than the ZigBee network as a function of the distance between the two nodes, which shows that ZigBee based networks can be used for longer communication range.

C. Throughput

Throughput can be expressed as the amount of data being sent or received during a defined time period. The aim of this test is to measure the throughput of both systems, as a function of the packet size. The first test was conducted for a point-to-point link using the "Throughput" tool in XCTU, the average throughput was measured over 20 s while varying the packet size between 10 bytes and 70 bytes. The experiment was then repeated for 2, 3, and 4 hops networks. During the tests, no route discoveries or failures occurred.

As expected and shown in Fig. 5, due to its larger frame size payload and its simpler address mode and packet header, DigiMesh network achieved better throughput than ZigBee based networks.

D. Packet delay

One significant indicator of a wireless mesh network performance is the packet delay, between two consecutive packets accurately received by the destination node without any error. In this experiment, the packet delay is a defined as the duration between sending a packet and receiving its acknowledgement at the source that is we define packet delay as the Round-Trip Time (RTT) until an ACK packet is received.

To measure packet delay for both systems, the software X-CTU was used. Using the frames generator functionality, a packet was created with the payload "Hello World", which has a size of 10 bytes, to be sent from the source node to the destination node. In total, 100 packets were sent to the destination node "A", where each packet was sent every 3 s. The times at which the packets were sent and the acknowledgement received were recorded using X-CTU, for both the ZigBee and DigiMesh network. The topologies studied in this experiment were point-to-point, and 2-, 3- and 4-hop networks. As shown in Fig. 6, ZigBee based networks outperformed DigiMesh no matter how many hopes between the transmitter

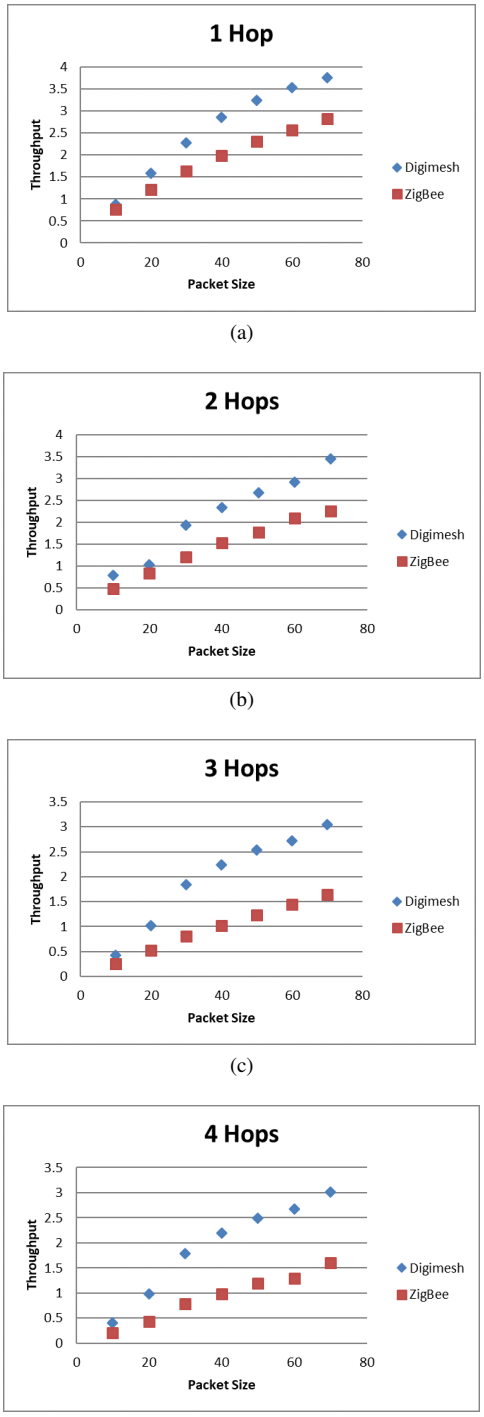

Fig. 5. Shows the throughput results as a function of the packet size for (a) 1, (b) 2, (c) 3 and (d) 4 hops' networks.

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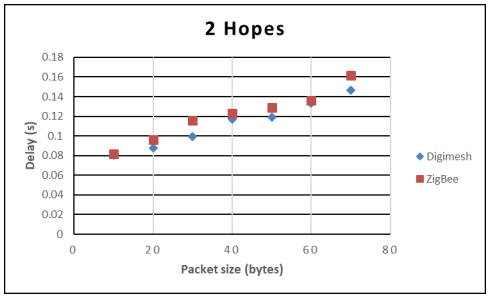
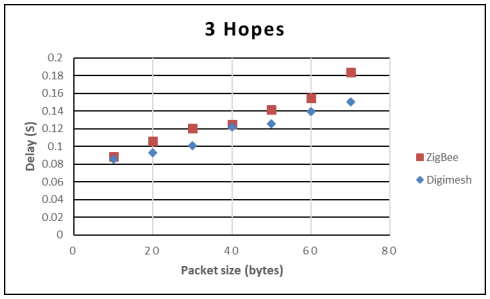


(a)



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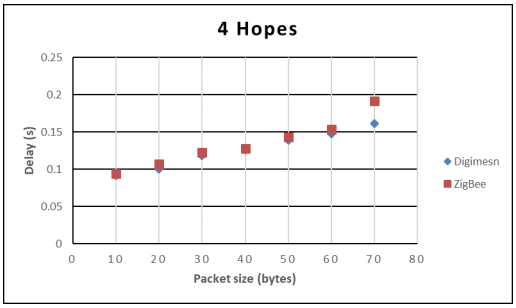

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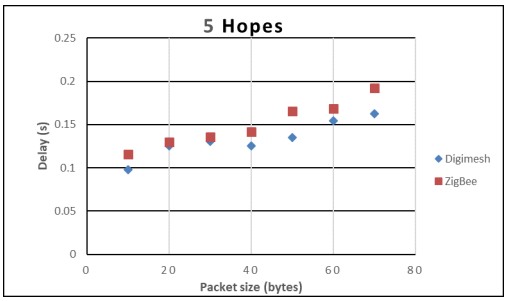

(d)

Fig. 6. Shows the RTT delay results as a function of the packet size for (a) 1 hope, (b) 2, (c) 3 and (d) 4 hops' networks.

and the receiver. This is due to the fact that ZigBee network has easier and more defined way to find the transmission path, due to the various nodes' types, for example, an End Device cannot send to another End Device, and should send its data to a parent node, which could be a router or coordinator node, this process optimizes the routing functionality and significantly reduces the transmission time, which results in having lower end to end RTT delay.

TABLE I
SHOWS THE MAXIMUM AND AVERAGE TIME TO FORM A NEW NETWORK FOR BOTH DIGIMESH AND ZIGBEE NETWORKS.

Criterion	DigiMesh	ZigBee
Maximum time to form new network	0.145 s	0.137 s
Average time to form new network	0.115 s	0.109 s

E. Mesh Routing Recovery Time

Both DigiMesh and ZigBee networks are characterized by being self-forming and self-healing networks, therefore no supervision is needed to create the network. After forming the mesh network, to relay messages from one device to another, the most optimized path is selected. However, if one of the routers is damaged or unable to communicate due to for instance, exhaustion of its battery, the network selects an alternative route. Hence, one of the most important characteristics of DigiMesh and ZigBee mesh networking is the self-healing ability. To test the time cost of mesh routing, the time between the elimination of one path and the selection of another was measured. The test was performed using the network topology illustrated in the figure below. Using XCTU packets containing the payload "Hello World" was sent from node "D" to node "A". In the DigiMesh network, node "A" is a router, and in the ZigBee network node "A" is the coordinator. The position of node "A" was fixed and node "D" was moved until the communication between the two nodes was lost for a considerable time. The two nodes "B" and "C" were added in the middle. In both the DigiMesh and ZigBee networks, nodes "B" and "C" are routers, and are acting as relays. After some time, power is cut off router "B", such that the new path is formed through router "C". Finally, the time between the last message sent through router "B" and the first message sent through router "C" was measured. The test was also repeated by powering off router "C". In total 20 trials were done for each network. As shown in Table I, the maximum time to find a new path measured in the case of DigiMesh network was 0.145 s, and the average was 0.115 s. In the case of the ZigBee network, the maximum time to find a new path was measured to be 0.137 s and the average time was 0.109 s. Therefore, the time to recover from a lost path is less for a ZigBee network.

V. CONCLUSION AND FUTURE WORKS

In this paper, an experimental performance analysis for both ZigBee and DigiMesh networks is presented. The experiments showed that DigiMesh networks have higher throughput. However, ZigBee networks outperformed DigiMesh in the RSSI as a function of the distance between adjacent nodes, the RTT delay and the time to form a new network after having a failure in the network. As a future work, the network robustness against external interference and noise will and the power consumption and network lifetime will be investigated for both types of networks.

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Acknowledgements:

This work is supported by the North Atlantic Treaty Organization (NATO) SPS Project No. SPS G4936, King Abdullah II Design and Development Bureau (KADDB), and the German Jordanian University.

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